

# LUNGANI ZUNGU

## Software Engineer

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## PROFESSIONAL SUMMARY

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Software Engineer at BMW Group South Africa with 3+ years building production cloud systems. I have independently delivered 9 live applications used daily on the plant floor - from AWS serverless workforce tools to Power BI dashboards and ML optimisation engines. BSc Computer Science & Statistics (78%, UKZN). PGDip Data Science in progress.

## KEY ACHIEVEMENTS

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**75% reduction in rotation planning time** - 2-3 hours per week reduced to 30 minutes through full-stack workforce management application

**100% elimination of training mismatches** - Zero incidents since production deployment of skills-aware rotation system

**20% reduction in rework cycle time** - Automated defect lookup via vehicle positioning (geofencing) and quality defect management system integration

**100% digital operations** - Eliminated all paper-based attendance, rotation, and headcount tracking

**>95% user adoption rate** - Achieved through structured training, GEMBA presence, and continuous support

**9 production systems delivered** - All actively used daily across shop floor, management, and enterprise teams

**AWS Hackathon 1st Place** - End-to-end analytics solution - data engineering, SQL analysis, and dashboard reporting

## WORK EXPERIENCE

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### BMW Group South Africa

March 2025 - Present

#### Software Engineer - Resource & Technology Steering

##### Full-Stack Application Development

- Designed and deployed the Work Organisation Toolset (WOT) - a full-stack production application built on React.js (frontend), AWS Lambda with Node.js (backend), API Gateway (REST APIs), DynamoDB (NoSQL database), S3 (document storage), and AWS Amplify (CI/CD hosting). Used daily by process supporters, process leaders, and shop floor associates.
- Delivered seven production modules within WOT: rotation planning with drag-and-drop scheduling, attendance management with automated email escalation, headcount management, skills matrix tracking (Level 1-4 with decay awareness), training management, relief management, and compliance reporting.
- Built a Rework Tracking System integrating vehicle positioning (geofencing) and quality defect management system data - automatically detecting vehicle entry into rework bays, pulling VIN-specific defects, filtering by work area, and recording rework session duration. Reduced defect lookup time from 5-10 minutes to zero and rework cycle time by 20%.
- Developing an ML-powered Intelligent Rotation Suggestion Engine using a custom scoring algorithm (decay risk, skill match, fairness, ergonomic safety constraints) to auto-generate optimised workforce rotation plans. Expected to reduce planning time from 30 minutes to under 10 minutes.

##### Cloud Data Engineering & AWS Migration

- Leading data engineering workstreams in enterprise AWS cloud migration - designing DynamoDB table structures, building Glue crawlers for data cataloguing, engineering end-to-end ETL pipelines, and establishing S3-based data lakes for centralised analytics storage.
- Building and deploying serverless data pipelines using AWS Lambda, Glue, Athena (serverless SQL on S3), and API Gateway, replacing on-premise systems with scalable cloud-native architecture.
- Rebuilding legacy dashboards in AWS QuickSight as part of the migration, integrating with Athena and DynamoDB data sources for real-time enterprise analytics.

##### Business Intelligence & Dashboard Development

- Built a Power BI Headcount Dashboard featuring waterfall charts, month-over-month variance, drill-down by port and shift, and one-click PowerPoint export - used weekly by management and HR Business Partners for workforce planning.

- Developed a Power BI Line Balancing Dashboard for process support planners, visualising average and maximum station loadings, KPI tracking, heat maps, and bottleneck identification - enabling 2-weeks-faster bottleneck resolution.
- Delivered a suite of Power BI Designer Dashboards connecting quality defect management system data to design ownership, enabling design engineers to track their components' shop floor performance and prioritise changes - reducing root cause analysis time by 50%.
- Built a Power BI Scrap Analysis Dashboard tracking scrap cost and volume by work package, defect type trends, and improvement ROI - used by management and quality teams for targeted corrective action.
- Designed Qlik Sense enterprise dashboards including associative data modelling, table design for optimal performance, load script optimisation, and operational metric visualisations - used enterprise-wide.

#### User Adoption & Change Management

- Implemented a structured training and adoption programme for all deployed applications - delivering hands-on training sessions with signed acknowledgments, one-page user guides, video tutorials, in-app tooltips, and regular GEMBA follow-up on the shop floor.
- Achieved greater than 95% daily active adoption in production environments with zero 'I wasn't trained' escalations - documented through signed acknowledgment records.
- Conducted regular GEMBA walks to gather user feedback, identify friction points, and deploy continuous improvements - maintaining trust and system relevance post-deployment.

#### MTN South Africa

*April 2023 - March 2025*

##### Data Scientist Graduate

- Performed quantitative and trend analysis on large financial and operational datasets using Python, SQL, SAS, and Power BI to support strategic decision-making.
- Built Power BI dashboards tracking revenue performance, cost efficiency, and customer behaviour KPIs for senior leadership.
- Delivered analysis contributing to a 25% sustained reduction in call centre volume through customer behaviour and process insights.
- Engaged with stakeholders across finance, marketing, and operations to define analytical requirements and present findings at leadership forums.
- Worked with financial data in a regulated telecoms environment, including revenue modelling and compliance-sensitive reporting.

#### EDUCATION

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##### University of KwaZulu-Natal

*2023 - Present*

##### Post Graduate Diploma in Data Science (In Progress)

##### University of KwaZulu-Natal

*2020 - 2022*

##### BSc in Computer Science and Statistics

Overall Average: 78%

#### TECHNICAL SKILLS

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**Frontend:** React.js, JavaScript (ES6+), HTML5, CSS3, responsive design, progressive web apps

**Backend:** AWS Lambda (Node.js), REST API design, API Gateway, serverless architecture

**Cloud & Data Engineering:** AWS (Lambda, DynamoDB, S3, Glue, Athena, QuickSight, API Gateway, Amplify), CI/CD (GitHub + Amplify), ETL pipeline design, data lake architecture

**Databases:** DynamoDB (NoSQL, data modelling), SQL (Oracle, Amazon Athena), database design and optimisation

**Business Intelligence:** Power BI (DAX, complex dashboards, waterfall charts), AWS QuickSight, Qlik Sense (associative data modelling, load script optimisation)

**Data Science & ML:** Python (pandas, numpy, scikit-learn), R, SAS, statistical modelling, predictive analytics, machine learning, scoring algorithms

**Manufacturing Systems:** vehicle positioning (geofencing), quality defect management system data integration, Lean Manufacturing, TIM WOODS, GEMBA, 5S, Line Balancing, Poka-Yoke

**Soft Skills:** User-centred design, change management, structured training delivery, GEMBA leadership, stakeholder communication, cross-functional collaboration

